C17

NE 763 001 – FALL 2026 – BEELER

Exam #3 (Paper)

Student's Name: Banerjee, Souri

Date of Exam: Monday, November 2, 2026 - Friday, November 13, 2026
Time Limit: ~~120 minutes~~ 180 minutes **(1.5x time)**
Appointment Start Time: 1:00pm, Friday, Nov 6, 2026
Appointment End Time: 4:00pm, Friday, Nov 6, 2026
Allowed Materials: Scientific Calculators
Student Accommodations: Extended Time (1.5T)

Academic Testing Services

Student – NC State University Pack Pledge

I, SOURI BANERJEE have neither given nor received unauthorized aid on this exam or assignment. I have read the instructions and acknowledge that this is the correct exam.

Student's Printed Name

Sour Banerjee 11/6/26

Student Signature *Date*

NOV 6 4:21

Exam must be turned in by: 4:19 SB

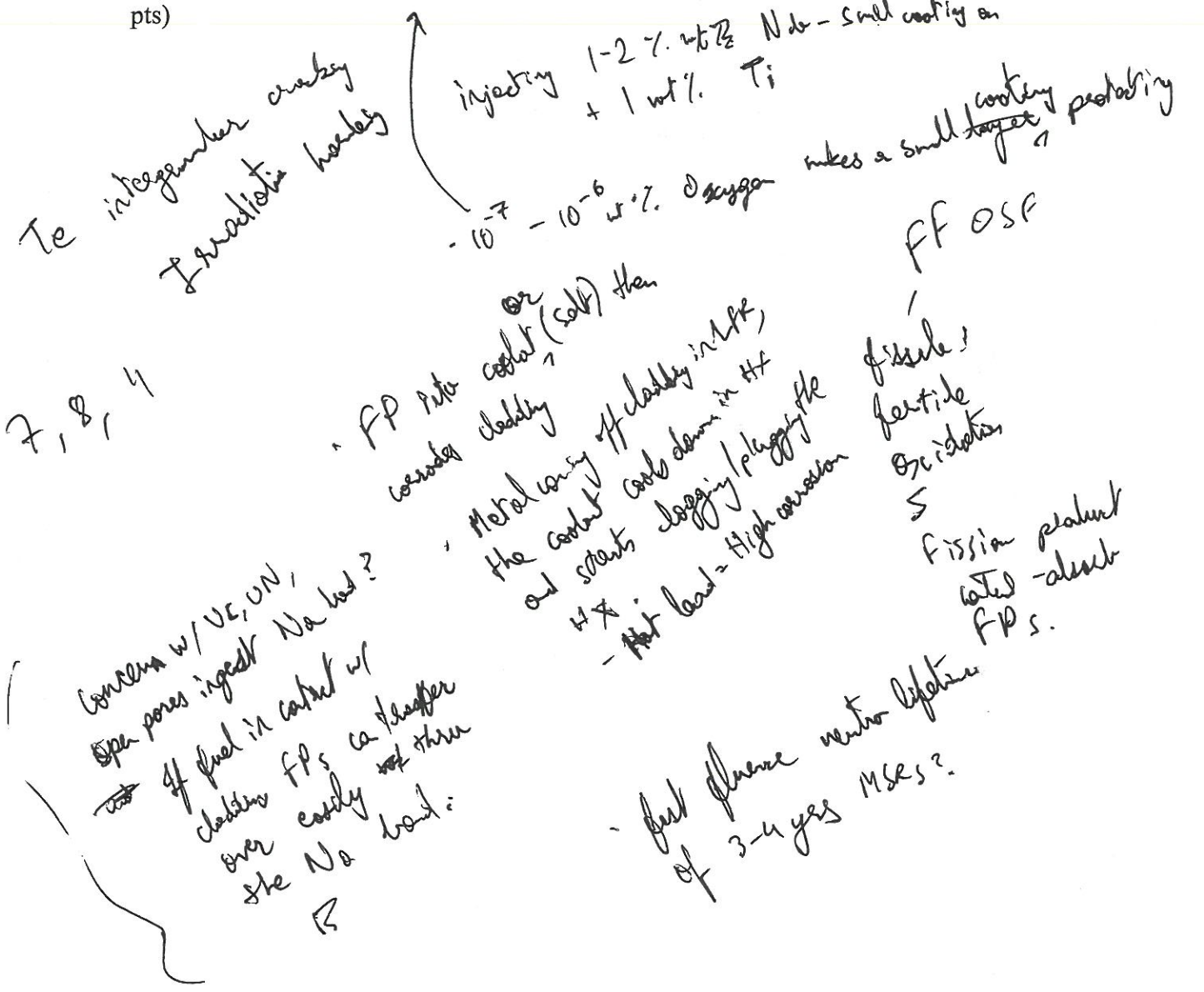
Exam End Time *Student's Initial Agreement*

NOTE: Failure to turn in exam on time may result in penalties at the instructor's discretion.

NE 763 Exam 3

Please provide your responses to the following questions. Point values indicated the expected depth of response.

1. What are the types and benefits of molten salt reactors? (6 pts)
2. What role do the different components of molten salt fuels serve? (6 pts)
3. What role does the cover/sparge gas serve in MSRE type reactors? (6 pts)
4. What does the UF4/UF3 ratio represent, and why do we care about it? (8 pts)
5. Discuss corrosion of the cladding in a molten salt environment. (16 pts)
6. Why are we interested in carbide and nitride fuels? (6 pts)
7. What are the two types of pin designs for C and N fuels? Discuss design and operational ramifications of the designs. (12 pts)
8. What are the three stages of temperature evolution in C and N fuels? (8 pts)
9. How do carbides and nitrides restructure as a function of burnup? (10 pts)
10. How do the C/M and N/M ratios change with burnup? Why is this important? How does this impact FCCI? (14 pts)
11. Why is there an oxygen window in lead-based coolant corrosion? Explain its impacts. (8 pts)



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1) ~~Benefits:~~ Types: • Solid fuel molten salt Rx — standard solid fuel w/ molten salt as coolant going around it transporting heat

• Static salt ~~Rx~~ MSR - Molten salt in cladding (the salt inside cladding does not move but is liquid), it use molten salts w/ good thermal conductivity as coolant

~~MSR~~ MCFR • Liquid molten salt Reactor: Uses liquid molten salt that flows around the primary loop generating heat, the secondary loop also is molten salt but w/ better thermal conductivity (for eg. FLiBe) (heat transferred between primary and secondary loop w/ HX). The tertiary loop is where for the SG (Steam generator), heat ~~transported~~ transferred from secondary to tertiary loop using another HX.

NOTE: in the solid fuel molten salt Rx, the fuel ~~is~~ can be TRISO pebbles.
in the static salt MSR, the salt inside cladding is the fuel salt.
in the Liquid molten salt Rx (loop type), the primary loop is ~~the~~ containing the fuel salt.

BENEFITS: The MSR operates at near atmospheric pressure, it's got great thermal conductivity and ~~is very efficient~~ can transfer heat efficiently for efficient electricity generation. Also, ~~it can't melt~~ because it's at near atmospheric operates at atmospheric pressure, the pressure vessel (PV) ~~needs to be~~ can have thinner walls and be smaller. It does not need to be able to handle higher pressures like ~15 MPa in PWRs. Thus it also makes the MSR cheaper. Fluoride has a strong negative reactivity coefficient reducing reactivity as the temp goes up. ~~and~~ Molten

4) The UF_4 / UF_3 ratio tells us about Redox. If the UF_4 / UF_3 ratio starts to increase and goes beyond 100, then ~~the~~ the metal fluoride / fluorine ratio will also start to drastically increase (MF_3 / F). ~~As the~~ These are for metals such as Cr, Ni, Mo, ~~Fe~~ and Fe. ~~As the metal fluoride / fluorine~~

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~~ratio increases~~, With increasing UF_4/UF_3 ratio the fluorine will start to eat away at the ~~salt~~ metal surfaces it comes into contact w/. Keeping an eye on UF_4/UF_3 ratio allows us to monitor the metal fluoride/fluorine ratio. We want to maintain a slightly reducing ratio so that the cladding and metal fluorides don't oxidize and corrode away.

2) The different components of the molten salt fuels control thermal conductivity, reactivity, ~~fuel~~ contain fissile material, fertile material, some corrosion control if possible, Burnable poisons, ~~BU~~ allows higher BU (burnup % of fuel), oxidation control, fission product control - absorb fission gases, etc.

3) The cover/sparge gas in MSRE can be Helium $(He)_2$. The He can prevent ~~IGA~~ or ~~to~~ reduce intergranular cracking caused by ~~Fe~~ Tellurium (Te Te is a fission product and enters the grain boundary (GrB) of the cladding/metal material. Having the He provides a small layer or barrier against the corrosive molten salt. It will protect the metal and cladding and any surface that would otherwise be exposed to the salt.

5) Corrosion Molten salts are really hard on the cladding surface and material when it comes to corrosion. Not only is the molten salt going to cause corrosion ~~on the surface~~ but the fission products (FPs) like Te that dissolve into the ~~salt~~ molten salt will start damaging the cladding. The Te will enter the GrBs of the cladding and start ~~microstructural~~ microscopic cracks on the GrB. The cladding ~~initially~~ ~~was~~ contains metals such as Cr, Mo, Ni, and Fe. ~~The salt~~ To prevent corrosion of cladding, engineers initially started off w/

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Hast-N, a nickel Based alloy, and then added 1-2 %. wt Nb and 1 wt % Ti. The Nb and Ti help keep Te at Tc^{2-} and form these fine precipitates which hold the Te in the grains and stop the Te from reaching the GB and cause intergranular cracking.

6] Carbide and nitride fuels (UC and UN) allow much improved thermal conductivity over oxides. They can reach considerably higher temps than Oxides as well as high linear power rates and higher BU allowing for the fuel to be used more efficiently. The carbides and nitrides can have a Na bond which will allow for the increased thermal conductivity.

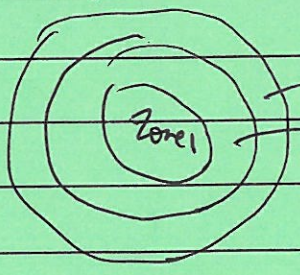
9) C and N (carbide and Nitride) ~~have go then the 3 temperature~~ restructure into 3 zones.

Zone 1: hottest zone in the center, the pores open up letting out fission gas.

Zone 2: intermediate region that experiences columnar grain growth (only for high linear power rates of >100 kW/m)

Zone 3: coolest outer region, maintains it's as-fab GB & microstructure

As BU increases, the temperature also increases w/ it.



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10) As BU ↑, the C/M and N/M ratios decreases. ~~This~~ is ~~important too because~~ as BU ↑, the ~~temps go up~~ The C/M and N/M ratio must be hyperstoichometric ~~Or~~ (greater than 1) given that C/M and N/M ratios decrease w/ BU increase. If C/M and N/M ratios go ~~to~~ below 1 they can cause ~~the~~ corrosion on the cladding via FCCI. ~~Additionally, given that the Na bond is present, it can~~ ~~The~~ Additionally, the lower than 1 C/M and N/M ratio ~~would~~ could lead to the fuel contacting the cladding. If the fuel contacts the cladding then the fission products can end up moving into the cladding ~~and~~ and causing corrosion. If a Na bond is present, it would ~~lead to~~ allow the FPs to be transported into the cladding material much easier.

11) The oxygen window is $10^{-7} - 10^{-6}$ w/ regards to lead based coolant corrosion. For background, lead is quite corrosive. Especially hot lead can corrode the ~~metal~~ piping material ~~as it~~ and carries the melted metal w/ it thru the piping system. ~~The~~ Once the Pb ~~is~~ or LBE (Pb-Bi Eutectic) reaches the ~~HX~~ it cools down and can end up clogging the piping system when it "~~drops~~ drops off" the metals that become solid once cooled. Now adding that ~~Oxygen~~ oxygen ~~about~~ a makes a small coating layer protecting the piping/cladding, or any material that comes into contact w/ the lead. However, if the oxygen is not enough then the protective coating will not be enough and will dissolve into the lead coolant. However, if the oxygen injected into system is higher than that range/window then the lead coolant ~~it~~ will start to oxidize ~~and will become a~~ sludge

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and will become a sludgy mess. We do not want corrosion or oxidation in our coolants, the purity is very important.

8) In C and N fuels (carbide, nitride), there are 3 stages of temperature evolution. The center of the fuel will be the hottest and it'll start the porosity will start increasing allowing fission product gases to leave the fuel. The intermediate region of the fuel will be a bit cooler and the grains will experience columnar grain growth but this columnar grain growth usually only occurs in cases where the linear power heat rate is quite high (> 100 kW/m).

The coolest region of the fuel is the outer region which maintains it's as-fab structure. Basically center = zone 1 (hottest), intermediate region = zone 2 (less hot), cool outer region = zone 3. The fuels would probably evolve from as-fab or as-sintered (when kinda cool), than it'll start experiencing some grain growth as it gets hotter (especially if it experiences high linear power heat rates), and then as it gets Consider hotter, the pores will open up.

7) I think the pin is voidal TRISO design could potentially count as 1 pin design. The other pin design is a solid slug in the shape of a cylinder. If there was a pin design that utilized a sodium bond, then that would be more simple to manufacture and assemble given that the fuel pin tolerances would be tight. However, the Na bond would give us cause for concern because it would easily transport the dissolved FPs

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from the fission process in the fuel to the cladding. These FPs are not friendly to the cladding material and would require additional monitoring and precautions and solutions that would reduce or mitigate the damage caused by ~~corrosion~~ corrosion by the FPs on the cladding. The solid cylinder slug w/o the sodium bond would need to be manufactured and assembled carefully, especially if it had tight tolerances which it would need to maintain good thermal conductivity ~~due to~~ (the fuel would need to touch the cladding). The contact between fuel and cladding would increase the chances for FCCI with the fission products ~~not~~ moving into the cladding surface and causing corrosion w/ the prismoidal fuel type, ~~when designing you would need to~~ the manufacturing and assembly of TRISO particles into the prism shape would be a bit challenging but doable as it was done in the past before. The prisms would then be assembled into a ~~round shape. circular pattern.~~ cylindrical pattern. After ~~that~~ the fuel is removed from Rx, it would need ~~to be different ways to be~~ efficient and cost effective ~~ways~~ ways for the fuel to be reprocessed because TRISO particles are challenging to break safely. Operationally, the sharp edges of the prisms would probably ~~be~~ be hot spots and can cause stress concentrations in the cladding.

4) Additional note: Reduction is chemical species gaining electrons, oxidation is the chemical species losing electrons.

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